

Colors in CSS

Learn how to define, apply, and control colors using common CSS color formats.

Why Colors Matter

Colors play a major role in the visual appearance of a website. In CSS, you can apply colors to text, backgrounds, borders, and other elements using different formats.

Ways to Define Colors

CSS supports several formats for defining colors:

1. Named Colors

CSS provides predefined color names such as **red**, **blue**, **green**, and **black**.

```
h1 {  
  color: red;  
}  
  
body {  
  background-color: #f0f0f0;  
}
```

2. HEX Codes

A hexadecimal value represents a color using a six-digit code. Shorthand can be used when each pair of hexadecimal digits is repeated.

```
#000000 -> black  
#ffffff -> white  
#ff0000 -> red  
  
#fff      -> same as #ffffff
```

3. RGB (Red, Green, Blue)

RGB defines a color using red, green, and blue channels. Each channel uses a value from **0 to 255**.

```
rgb(0, 0, 0)           -> black
rgb(255, 255, 255)    -> white

p {
  color: rgb(255, 0, 0);
}
```

4. RGBA (RGB + Alpha)

RGBA extends RGB with an alpha value that controls transparency. An alpha value of **0** is fully transparent, while **1** is fully opaque.

```
div {
  background-color: rgba(0, 0, 0, 0.5);
}
```

This creates a semi-transparent black background.

5. HSL (Hue, Saturation, Lightness)

HSL defines colors using three components:

Component	Meaning
Hue	The color angle on the color wheel.
Saturation	The intensity of the color.
Lightness	The brightness of the color.

```
h2 {
  color: hsl(240, 100%, 50%);
}
```

6. HSLA (HSL + Alpha)

HSLA works like HSL but adds an alpha value for transparency.

```
section {
  background-color: hsla(120, 60%, 70%, 0.3);
}
```

Applying Colors in CSS

Color-related properties can be used in many places, including text, backgrounds, and borders.

```
h1 {  
  color: navy;           /* Text color */  
  background-color: #e0e0e0; /* Background color */  
  border: 2px solid #333;  /* Border color */  
}
```

Transparent and currentColor

Value	Purpose
transparent	Makes an element's color fully transparent.
currentColor	Uses the current value of the element's color property.

```
button {  
  color: blue;  
  border: 2px solid currentColor;  
}
```

Color Format Quick Reference

Format	Example	Key Idea
Named	red	Predefined CSS color name.
HEX	#ff0000	Hexadecimal color notation.
RGB	rgb(255, 0, 0)	Red, green, and blue channels from 0-255.
RGBA	rgba(0, 0, 0, 0.5)	RGB with an alpha value for transparency.
HSL	hsl(240, 100%, 50%)	Hue, saturation, and lightness.
HSLA	hsla(120, 60%, 70%, 0.3)	HSL with an alpha value for transparency.

Tip

- Use color to enhance readability, structure, and visual aesthetics.
- Choose the format that best suits your workflow.
- Use RGBA or HSLA when you need transparency effects.

Quick Takeaway

CSS provides multiple ways to define colors. Named colors are simple, HEX is compact, RGB and HSL provide structured color values, and RGBA and HSLA add transparency control.